Decision Log – Business Intelligence Agent (Monday.com NLP System)

1. Key Assumptions

Several assumptions were made while designing and implementing the Business Intelligence Agent in order to simplify development and ensure the system could operate reliably within the available time and technical constraints.

1. Monday.com as the single source of truth

The system assumes that Monday.com boards contain the primary operational data for the organization. Specifically, it assumes that:

- Deals, work orders, or operational tasks are stored in Monday boards.
- Each board item represents a business record (deal, order, or activity).
- Important business fields such as **revenue, sector, stage, status, and dates** exist as column values.

Because Monday.com stores column values in a semi-structured JSON format, the system assumes that these values can be normalized into structured data using Python.

2. Limited schema variability

Another assumption is that the schema of the boards will remain relatively stable. While the code dynamically reads column values, it still expects recognizable fields such as:

- revenue / value
- sector / category
- status
- created date or close date

If board structures change significantly, additional mapping logic may be required.

3. Natural language queries are relatively simple

The NLP system assumes that users will ask **short and direct business questions**, such as:

- "What is revenue this month?"
- "Show deals in healthcare sector."
- "How many deals were won last week?"

The system does not assume complex grammar, long paragraphs, or multi-step reasoning. Instead, it relies on **keyword detection and entity extraction**.

4. Data size is moderate

The implementation uses **pandas DataFrames** for analysis. Therefore, it assumes the dataset is small enough to be handled in memory. This is reasonable for most Monday.com boards used by small or medium organizations.

5. Users are business stakeholders

The design assumes the end users are **non-technical leadership or operations staff** who prefer asking questions in natural language rather than writing SQL queries or using complex dashboards.

2. Trade-offs Chosen and Why

Several architectural and technical trade-offs were made to balance **simplicity, reliability, and development speed**.

Trade-off 1: Rule-Based NLP vs Machine Learning Models

The system uses a **rule-based NLP pipeline** instead of machine learning models such as transformers or large language models.

Advantages:

- No dependency on external AI APIs
- Fully deterministic outputs
- Faster execution
- No model training required
- Easier debugging

Disadvantages:

- Limited language understanding
- Less flexible to complex queries
- Requires manual keyword mapping

This decision was made because the goal was to build a **lightweight and reliable BI assistant** rather than a full conversational AI system.

Trade-off 2: Pandas Instead of Database Query Engines

The system stores the cleaned data inside **pandas DataFrames** instead of a database such as PostgreSQL.

Advantages:

- Very fast development
- Simple aggregation functions
- Ideal for small-to-medium datasets
- No additional infrastructure required

Disadvantages:

- Not suitable for very large datasets
- No persistent storage
- Limited multi-user scalability

Since the dataset retrieved from Monday.com boards is typically not extremely large, pandas provides a **good balance between simplicity and analytical power**.

Trade-off 3: Notebook-Based Implementation

The project is implemented primarily inside a **Jupyter Notebook**.

Advantages:

- Easy experimentation
- Clear step-by-step execution
- Suitable for demonstrations
- Good for educational purposes

Disadvantages:

- Not ideal for production deployment
- Harder to maintain as a large application

This format was chosen because the project is intended as a **prototype BI assistant** rather than a fully deployed product.

Trade-off 4: Direct API Calls vs SDK Libraries

The system directly calls the **Monday.com GraphQL API using requests** instead of using a third-party SDK.

Advantages:

- Full control over queries
- Fewer dependencies
- Better understanding of API behavior

Disadvantages:

- Requires manual request construction
- Slightly more code

However, this approach ensures the system remains **lightweight and transparent**.

3. What I Would Do Differently With More Time

If more development time were available, several improvements could significantly enhance the system.

1. Add a conversational NLP model

The current rule-based NLP system could be replaced or augmented with a **machine learning-based intent classifier** or transformer-based NLP model. This would allow:

- understanding longer questions
- improved entity extraction
- better context handling
- conversational follow-up queries

2. Build a web interface

Currently, queries are executed within the notebook environment. A **web dashboard** using frameworks such as:

- Streamlit
- Flask
- FastAPI

could provide a user-friendly interface where leadership can simply type questions and see results instantly.

3. Implement real-time dashboards

Another improvement would be integrating **interactive dashboards** that automatically update metrics such as:

- revenue trends
- deal pipelines
- sector performance
- weekly business KPIs

Libraries like **Plotly or Dash** could support this.

4. Improve data normalization

Monday.com column values can be inconsistent depending on board configuration. More time would allow development of a **robust schema mapping layer** to automatically standardize:

- status values
- currency formats
- date formats
- categorical fields

5. Add predictive analytics

Currently, the system only performs descriptive analytics. With more time, the system could include:

- revenue forecasting
- deal win probability prediction
- anomaly detection
- growth trend analysis

This would transform the system from **business intelligence to decision intelligence**.

4. Interpretation of “Leadership Updates”

The phrase **"leadership updates"** was interpreted as **concise summaries of key operational and financial metrics that executives need for decision-making**.

Leadership typically does not require raw data tables. Instead, they need **clear, high-level insights** such as:

- revenue performance
- number of deals closed
- sector performance
- operational workload
- pipeline opportunities

Therefore, the system generates **short narrative summaries** that highlight:

- key KPIs
- top-performing sectors
- deal outcomes
- notable trends

An example leadership update generated by the system might include:

- total revenue during a period
- number of deals won
- number of deals lost
- top-performing sector
- number of open opportunities

The goal is to replicate the type of **weekly or monthly operational summary** that leadership teams review during management meetings.

This interpretation emphasizes **clarity, brevity, and actionable insights** rather than raw analytics.

Conclusion

The Business Intelligence Agent was designed as a **lightweight, explainable, and practical solution for querying operational data using natural language**. By combining Monday.com API integration, data normalization, pandas-based analytics, and a rule-based NLP pipeline, the system demonstrates how business users can interact with organizational data more intuitively.

Although the current implementation prioritizes simplicity and transparency, the architecture allows future expansion into more advanced areas such as conversational AI, predictive analytics, and real-time dashboards.